Experiment No.: 5

Date of Performance :
Date of Submission :

Aim: Measurement of signal strength vs distance for Wi-Fi router.

Required Equipment / Software :

- Wi-Fi router/access point with a stable wireless signal.
- Android smartphone or tablet with Wi-Fi capability.
- Wi-Fi analyzer application such as WiFi Analyzer/WiFi man or another suitable Wi-Fi signal monitoring app.

Theory :

Wi-Fi communication uses radio waves to transfer data between a wireless router/access point and client devices. The received signal generally becomes weaker as the distance between the transmitter and receiver increases.

Received Signal Strength Indicator (RSSI) is commonly used to represent the received wireless signal level. Wi-Fi analyzer applications usually display RSSI in dBm. RSSI values are normally negative; a value closer to 0 dBm represents a stronger signal. For example, −45 dBm is stronger than −75 dBm.

In an ideal free-space environment, received power decreases approximately with the square of distance. In practical indoor Wi-Fi environments, walls, furniture, human bodies, antenna orientation, multipath propagation and interference affect the measured RSSI.

A simplified path-loss relationship is: $RSSI(d) \approx RSSI(d_0) - 10n \log_{10}(d/d_0)$, where d is the measurement distance, d_0 is a reference distance and n is the path-loss exponent.

Procedure :

1. Place the Wi-Fi router at a fixed location and keep its position and orientation

unchanged.

2. Connect the smartphone to the selected Wi-Fi network.

3. Open a suitable Wi-Fi analyzer/WiFi man signal monitoring application.

4. Note the SSID, Wi-Fi band and channel of the selected router.

5. Measure and mark distances such as 1 m, 2 m, 3 m, 5 m, 7 m and 10 m from the router.

6. At each distance, keep the smartphone at approximately the same height and orientation.

7. Wait for a few seconds for the RSSI reading to become reasonably stable.

8. Record the RSSI value in dBm and signal quality in the observation table.

9. Repeat the measurement for all selected distances.

10. Compare the RSSI values and observe the general relationship between distance and signal strength.

11. Plot Distance (m) on the X-axis and RSSI (dBm) on the Y-axis, if required.

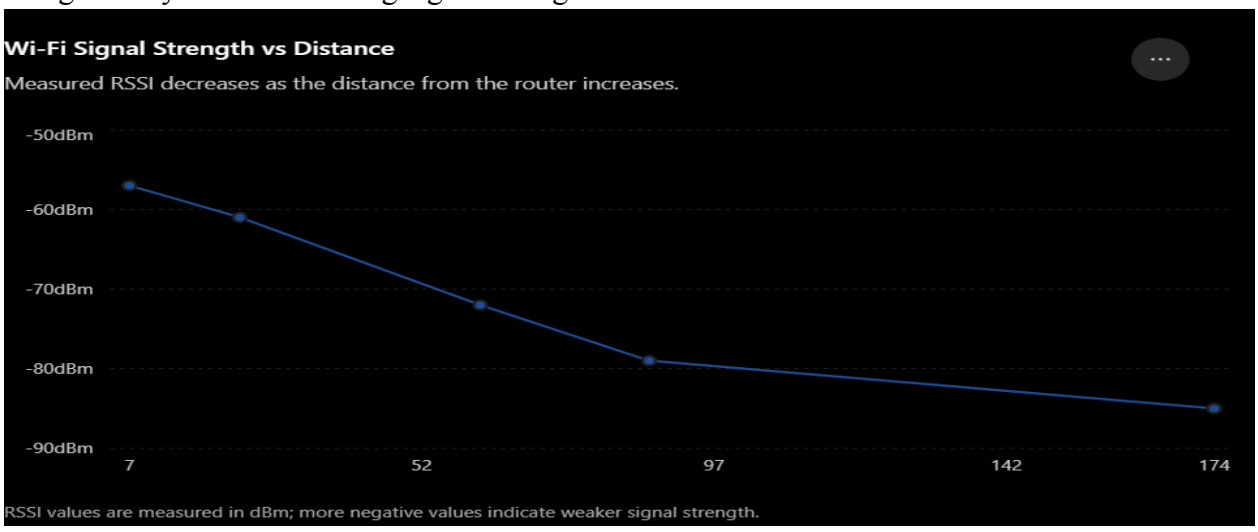
12. Explain any unusual readings using obstacles, reflections, multipath propagation or interference.

Observation Table :

Sr. No.	SSID	Distance (m)	Wi-Fi Band	Channel No.	RSSI (dBm)	Signal Quality
1	Ruhaani_4G	~174	2.4	6	-85	Very Weak
2	Atharvaj_4G	~7	2.4	1	-57	Good
3	Airtel_R k bharate	~24	2.4	11	-61	Good
4	Airtel_22	~87	2.4	6	-79	Weak
5	TP_Link_9A2211	~61	2.4	11	-72	Moderate

Approximate signal-quality guide: Very Strong (−30 to −50 dBm), Good (−51 to −67 dBm), Moderate (−68 to −75 dBm), Weak (−76 to −85 dBm), Very Weak (below −85 dBm). These ranges are approximate.

Graph : Plot Distance from Router (m) on the X-axis and RSSI (dBm) on the Y-axis. The graph will generally show decreasing signal strength as distance increases.



Result :

Wi-Fi signal strength was successfully measured at different distances from the router using a mobile Wi-Fi analyzer. The observations generally show that the received signal becomes weaker as distance increases.

Conclusion:

The experiment demonstrates the relationship between Wi-Fi signal strength and distance. In general, increasing the distance between the router and client reduces received signal strength. Practical readings are influenced by obstacles, reflections, interference and antenna orientation.

Viva Voce Questions :

1. What is RSSI?

2. Which is stronger, −40 dBm or −70 dBm?

3. What happens to Wi-Fi signal strength as distance increases?

4. Why is RSSI usually negative?

5.Why may RSSI not decrease smoothly indoors?

MARKS & SIGNATURE :

R1 (3 Marks)	R2 (5 Marks)	R3 (4 Marks)	R4 (3 Marks)	Total (15 Marks)	Signature